

CAREER OPPORTUNITIES ACROSS INDUSTRIES

For engineering aspirants, here is an overview of the key skills required and major recruiters in varied industries to help aspirants decide the best career option for them

Aerospace Engineering

Aerospace engineering deals with the design, construction, development, testing and manufacture of aircrafts, spacecrafts, missiles, military planes, space shuttles, satellites and hover-crafts—in other words, vehicles/systems that are meant for atmospheric and space environments. Aerospace engineers work within a team of other engineers to develop new technologies for use in aviation, defence systems and space exploration, often specialising in such areas as structural design, guidance, navigation and control, instrumentation and communication, or production methods.

Key skills required

Although experienced applicants are always preferred by the companies in aerospace, recruiters, while hiring a fresher, usually look for the technical and problem-solving skills of the student. In general, core companies look for the basic knowledge and understanding of the subject matter and IT companies look for coding skills and aptitude.

While hiring experienced professionals, companies look for people who are experts in tackling problems related to one or more of the following fields: computational fluid dynamics, aerodynamics, computational structural analysis, aircraft structures, control and guidance, and propulsion systems.

Who's hiring?

Government/public sector organisations like ISRO, DRDO, HAL, Air India and National Laboratories; private-sector engineering and IT companies like Ashok Leyland, Godrej, TCS, Infosys, Wipro and Ramco; MNCs like GE, Honeywell, Safran, Cessna, Volvo, Airbus, and Boeing-specialised software companies like Ansys & Fluent, Fluidyn and Cd-Adapco

Quality Testing

From personal computers to home appliances, car components and industrial electronics, quality testing is part of every manufacturing business. Before any product is sold, it is required to undergo testing and quality control. It is the quality testing professionals who carry out these tests and approve of a product's compliance to set quality standards.

Quality testing professionals are required to constantly analyse the data available to them, and compile test results to make recommendations to product developers and quality assurance professionals in order to better the products.

Key skills required

Candidates who wish to pursue quality testing as profession should have a quantitative, analytical aptitude. They should be able to reason mathematical problems, and, at the same time, must be observant to pick a quality related issue and brainstorm over it to find its fix.

People who wish to pursue it also need to be technically sound and equipped with a host of knowledge on test and measurement equipment, failure mode and effects analysis (FMEA), data analysis methods and engineering models.

The field is not restricted to the technical aspects of testing and quality management alone. Managerial skills are as important as technical qualifications.

Who is hiring?

Experts believe that aerospace and medical sectors are going to be hot grounds for quality testing professionals in the near future. As the Indian electronics manufacturing industry is becoming obsessed with quality products that could compete globally, proficient quality testing professionals have much to look forward to in the years to come.

Based on their products and processes, companies have specific requirements for quality testing professionals and hire accordingly. One must have full knowledge of the process and product, as well as quality tools.